Retail Insights

Superstore Sales Analysis — Full Project Documentation

Python · MySQL · Power BI Desktop

9,977 Orders · $2.30M Sales · $286.4K Profit · 12.47% Margin

TOOLS	SOURCE	RECORDS	SCOPE
Python · MySQL · Power BI	Superstore Retail Dataset	9,977 · 12 Cols	United States, 4 Regions

01 EXECUTIVE SUMMARY

- **Cleaned and analyzed 9,994 raw US retail transactions** down to a 9,977-row, 12-column analysis-ready dataset — removing 17 exact duplicates and 1 constant column (Country), through a fully scripted, repeatable Python pipeline.
- **Answered 9 structured business questions** covering profitability, category/sub-category performance, discounting behavior, regional strength, customer segments, and fulfillment — then reproduced every answer independently in MySQL, including a hand-built Pearson correlation formula for the discount-profit relationship.
- **Built a 2-page, 15-visual interactive Power BI dashboard** with 5 DAX measures, a red/blue conditional-formatted profit-by-sub-category chart, a state-level filled map, and cross-filtering region/category slicers — connected directly to MySQL.
- **Surfaced a critical profitability gap hiding inside a strong top-line:** Furniture is the 2nd-highest category by sales ($741,306) yet dead last by profit ($18,422, a 2.5% margin) — and Tables specifically lose $17,725 outright. Meanwhile Technology leads on both sales and profit, and Discount is negatively correlated with Profit (r = -0.22), directly explaining where the margin is leaking.

02 PROBLEM STATEMENT

The Superstore dataset arrived as a flat transactional export — 9,994 line items across shipping, customer segment, geography, product category, and financials — with no existing reporting layer. On the surface, aggregate performance looked healthy ($2.3M in sales), but leadership had no visibility into which categories, sub-categories, regions, or discount levels were actually generating profit versus quietly eroding it. The goal of this project was to build a clean, validated pipeline — Python for exploratory analysis, SQL for relational querying, and Power BI for an interactive dashboard — that could answer 9 concrete profitability questions and surface exactly where the business was winning and losing money.

03 OBJECTIVES

- **Obj 1:** Diagnose data-quality issues (duplicates, constant columns, encoding) before writing any cleaning code.
- **Obj 2:** Clean the dataset in a scripted Python pipeline and answer all 9 project questions with summary statistics and visualizations.

- **Obj 3:** Load the cleaned data into MySQL and reproduce every Python finding as an independent SQL query.
- **Obj 4:** Build DAX measures and a 2-page interactive Power BI dashboard connected live to MySQL, including a geographic view.
- **Obj 5:** Identify specific loss-making sub-categories and the discounting behavior driving them, not just top-line totals.
- **Obj 6:** Produce a portfolio-ready deliverable demonstrating the full pipeline with one consistent, cross-validated source of truth.

04 DATASET OVERVIEW

Source & Structure

Attribute	Detail
Dataset Name	Superstore Retail Sales Dataset
File Format	CSV (latin1 encoding — contains non-UTF8 city/state characters)
Raw Records	9,994 rows × 13 columns
Clean Records	9,977 rows × 12 columns (after dedup + column drop)
Geographic Scope	United States only · 49 states · 531 cities · 4 regions
Duplicate Rows Removed	17 exact duplicates
Columns Dropped	Country (constant value 'United States' — zero analytical use)
Missing Values	Zero across all 13 original columns

Key Columns Used in Analysis

Column	Type	Description
category	Text	3 top-level categories: Furniture, Office Supplies, Technology
sub_category	Text	17 product sub-categories (e.g. Tables, Chairs, Copiers)
region	Text	4 US regions: West, East, Central, South
segment	Text	3 customer segments: Consumer, Corporate, Home Office
ship_mode	Text	4 modes: Standard Class, Second Class, First Class, Same Day
sales	Decimal	Line-item revenue in USD (total $2,297,200.86)
profit	Decimal	Line-item profit in USD — can be negative (total $286,397.02)
discount	Decimal	Discount rate applied, 0.00–0.80 (0–80%)

05 TOOLS & TECHNOLOGIES USED

Tool	Platform	Purpose in This Project
Python 3	VS Code	Data cleaning, EDA, statistics, matplotlib/seaborn visualization
pandas / numpy	Python library	Duplicate removal, groupby aggregation, correlation analysis

Tool	Platform	Purpose in This Project
MySQL 8	MySQL Workbench	Relational storage of cleaned data; SQL validation of all 9 answers
Power BI Desktop	Microsoft	2-page interactive dashboard, DAX measures, filled map, slicers
DAX	Power BI native language	5 measures: Total Sales, Total Profit, Profit Margin %, Total Orders, Avg Discount

06 DATA CLEANING PROCESS — PYTHON

Only 2 real issues existed in this dataset — both fixed in a 4-step scripted pipeline.

Step	Action	Result
1	Load CSV with encoding='latin1'	9,994 rows × 13 columns
2	Drop 17 exact duplicate rows	-17 rows → 9,977 remain
3	Drop Country column (constant, zero variance)	13 → 12 columns
4	Standardize column names (spaces/hyphens → underscores)	MySQL-safe column names

07 SQL / MYSQL — VALIDATION LAYER

A single flat table (orders), imported via LOAD DATA LOCAL INFILE, with 4 indexes on the columns used in every GROUP BY (category, sub_category, region, segment). All 9 Python answers were reproduced independently in MySQL.

The Correlation Problem

MySQL has no built-in CORR() aggregate. The discount-profit relationship (Q4) was implemented manually using the Pearson formula: $r = (n\Sigma xy - \Sigma x\Sigma y) / \sqrt{[(n\Sigma x^2 - (\Sigma x)^2)(n\Sigma y^2 - (\Sigma y)^2)]}$. Result: -0.2197, confirming a real (if moderate) negative relationship between discount depth and profit.

Analytical VIEW

A vw_superstore_summary view pre-aggregates orders into discount buckets (No Discount / Up to 20% / 20-40% / Over 40%) crossed with region, category, sub_category, segment, and ship_mode — enabling a VIEW Weighted Margin % measure in Power BI without rescanning all 9,977 rows per query.

08 DAX MEASURES CREATED

Measure	DAX Formula	Result
Total Sales	SUM(orders[sales])	$2,297,200.86
Total Profit	SUM(orders[profit])	$286,397.02
Profit Margin %	DIVIDE([Total Profit],[Total Sales])*100	12.47
Total Orders	COUNTROWS(orders)	9,977
Avg Discount	AVERAGE(orders[discount])	~0.16
VIEW Weighted Margin %	SUMX-weighted average from vw_superstore_summary	12.47 (cross-check)

Formatting note: Profit Margin % already multiplies by 100 in the DAX formula. Format the card as plain Number (2 decimals) with a manual "%" suffix, not as Power BI's Percentage type — Percentage format would multiply by 100 a second time (12.47 → 1,247%).

09 DASHBOARD VISUALS BUILT (15 Visuals, 2 Pages)

Page 1 — Sales & Profit Overview

#	Visual Type	Fields Used	Insight Delivered
1-3	KPI Cards	Total Sales / Total Profit / Profit Margin %	$2.30M sales, $286.4K profit, 12.47% margin
4	Bar Chart	Y: category / X: Total Profit	Furniture trails badly despite 2nd-highest sales
5	Bar Chart (conditional color)	Y: sub_category / X: Total Profit	Tables and Bookcases render in red — the only loss-makers
6	Filled Map	Location: state / Color: Total Sales	Geographic concentration across 49 states
7-8	Slicers	region, category	Cross-filters all Page 1 visuals

Page 2 — Discounting & Fulfillment

#	Visual Type	Fields Used	Insight Delivered
1	Scatter + trend line	X: discount / Y: profit (Don't summarize)	Visible downward slope, corr = -0.22
2	Donut Chart	Legend: segment / Values: Total Sales	Consumer segment dominates at $1.16M
3	Clustered Column	X: ship_mode / Y: Total Orders	Standard Class handles 60% of all orders
4	Table	quantity, Avg Profit	Order-size vs. profitability relationship
5	Slicer	ship_mode	Cross-filters Page 2 visuals

10 KEY INSIGHTS & RESULTS

A. Overall Profitability

- **The business is profitable but thin:** $2,297,200.86 in sales converts to $286,397.02 in profit — a 12.47% overall margin.

B. Category & Sub-Category Performance

- **Technology is the strongest category on every metric:** $836,154 in sales and $145,455 in profit — the clear leader on both.
- **Furniture is the hidden problem category:** $741,306 in sales (2nd highest) but only $18,422 in profit (last place, 2.5% margin) — a category that looks healthy on a sales report but isn't.
- **Two sub-categories are outright unprofitable:** Tables lose $17,725 and Bookcases lose $3,473 — both should be flagged for pricing or discount-policy review.

C. Discounting Behavior

- **Discount and Profit are negatively correlated** (r = -0.2197) — heavier discounting is measurably associated with thinner (or negative) profit, a direct, quantified link between the discounting policy and the Furniture/Tables problem above.

D. Regional & Segment Performance

- **West leads all regions** at $725K in sales, followed by East ($679K), Central ($501K), and South ($392K).
- **Consumer is the most valuable segment** on both sales ($1,160,833) and profit ($134,007); Home Office is smallest on both ($429,293 / $60,279).
- **Standard Class dominates fulfillment** with 5,968 of 9,977 orders (59.8%) and $1,358,216 in sales — nearly 4x the next-highest mode.

11 CHALLENGES FACED & HOW OVERCOME

Challenge 1 — Non-UTF8 Characters in Source Data

Problem: Standard pd.read_csv() with default encoding failed on several city/state name characters.

Solution: Loaded with encoding='latin1', which correctly parsed all rows without silently dropping or corrupting any location data.

Challenge 2 — MySQL Has No Native Correlation Function

Problem: The discount-profit relationship (Q4) has no direct SQL equivalent to pandas' .corr() in MySQL.

Solution: Implemented the Pearson correlation formula manually using SUM(), POWER(), and SQRT() — validated the SQL output (-0.2197) against the Python result to 4 decimal places.

Challenge 3 — A Constant Column Hiding in the Schema

Problem: The Country column added a full VARCHAR field to every row while providing zero analytical value (every row was 'United States').

Solution: Checked .nunique() on every column during the assessment phase before cleaning — caught and dropped it rather than carrying dead weight through the whole pipeline.

Challenge 4 — A Profitable-Looking Category Masking a Loss-Making Sub-Category

Problem: A category-level view alone (Furniture: $741K sales) would have missed that Tables specifically loses $17,725 — the aggregate was hiding the real problem.

Solution: Always paired category-level and sub-category-level breakdowns rather than stopping at the first level of aggregation — the sub-category chart is what actually surfaces the actionable finding.

12 SKILLS LEARNED & TAKEAWAYS

- End-to-end pipeline design: raw CSV → Python cleaning/EDA → relational SQL validation → interactive BI dashboard, with consistent numbers validated at every layer.
- Encoding awareness: recognizing and handling non-UTF8 source data instead of letting a default read silently fail or corrupt values.
- Multi-level aggregation discipline: category-level totals alone can hide sub-category-level losses — always drill one level deeper before concluding.

- DAX fundamentals: DIVIDE() for safe division, and the specific formatting trap of double-applying a percentage multiplier.
- Conditional formatting as an analytical tool, not just decoration: color-coding negative-profit sub-categories makes the loss-makers immediately visible without reading every value.
- Translating pandas groupby logic into equivalent SQL, including a hand-built Pearson correlation formula for a function MySQL doesn't provide natively.

13 CONCLUSION

- **Processed 9,994 raw retail transactions** through a 4-step Python cleaning pipeline, reaching a 9,977-row, 12-column analysis-ready dataset with zero blind cleaning decisions.
- **Independently validated all 9 project questions in MySQL,** including a manually-implemented Pearson correlation formula for the discount-profit relationship, confirming agreement with Python to 4 decimal places.
- **Delivered a 2-page, 15-visual Power BI dashboard** connected live to MySQL, with 5 DAX measures, a conditional-formatted loss-maker chart, and a state-level map.
- **Most importantly, surfaced an actionable business finding hiding beneath a healthy top line:** Furniture's high sales mask a near-zero margin, Tables loses money outright, and heavier discounting is measurably linked to that erosion — a direct, three-part causal story the data supports end to end.

KPI Summary

Metric	Value	Metric	Value
Total Orders	9,977	States / Cities Covered	49 / 531
Total Sales	$2,297,200.86	Total Profit	$286,397.02
Overall Margin	12.47%	Discount ↔ Profit Corr.	-0.2197
Top Category (Profit)	Technology — $145,455	Weakest Category (Profit)	Furniture — $18,422
Biggest Loss-Maker	Tables — -$17,725	Top Region	West — $725K
Top Segment	Consumer — $1.16M sales	Dominant Ship Mode	Standard Class — 59.8%
Duplicates Removed	17	Columns Dropped	1 (Country)